Micro II Study Guide

Pathology microteaching · B12 list, 31 slides · one key picture per case. ★ = tested in 3 or more past years · M3 = asked in the Micro III exam that year.

Approach to any tumor

1. **Border:** circumscribed or infiltrative; in situ or invasive; how deep.
2. **Cytology:** N/C ratio, hyperchromasia, pleomorphism, nucleoli, mitoses, necrosis.
3. **Architecture:** sheets, nests, glands; fascicles, whorls, palisading.
4. **Cell lineage:** keratin and bridges (squamous), glands and mucin (adeno), cigar nuclei (smooth muscle), wavy nuclei (nerve sheath).
5. **Stroma:** desmoplasia, myxoid, collagenous, lymphoid.

Write the description in this order. Past graders scored exactly these aspects (year 10).

Scoring

- Each question: organ 2 + diagnosis 5 + description ~10.
- Key diagnostic finding first; it carries the most points.
- Findings not on the slide cost points. Benign tumors: write “no cytologic atypia”.
- No abbreviations in the diagnosis.

Highest yield (07–11)

- **Squamous cell carcinoma:** 4 years in Micro II; esophagus, cervix, skin.
- **Gastric signet-ring carcinoma** and **NET:** 4 years each (some in Micro III).
- **Atherosclerosis, Wilms tumor, colon adenocarcinoma, amyloid:** 3 years each.
- **Schwannoma, mature teratoma:** 2 years each.

Organs changed in past exams

- SCC → skin (11) · amyloid → salivary gland (11)
- Neurofibroma → skin (10) · melanoma → lung (10)
- NET → pancreas (10), duodenum (11)

Name the organ you see, not the one you memorised.

Slide numbers that differ

- Hyaline membrane disease: PA0200 (was PA0341)
- Gastric intestinal type: lecture PA0049; list PA0309
- Atherosclerosis: lecture PA0170; list PA0340
- Acute cellular rejection: PA0061 and PA0269

Comparison tables

Small round blue cell tumors

	Neuroblastoma	Retinoblastoma	Wilms tumor
Site	Adrenal medulla, sympathetic ganglia	Retina	Kidney
Key	Neuropil; Homer-Wright rosettes	Flexner-Wintersteiner rosettes; fleurettes	Triphasic: blastema, epithelium, stroma
Age	Median 18 months	About 24 months	Peak 2–5 years
Gene	MYCN, ALK	RB (two hits)	WT1, WT2

Rosettes

Type	Centre	Seen in
Homer-Wright	Neuropil	Neuroblastoma, retinoblastoma
Flexner-Wintersteiner	Lumen	Retinoblastoma
Perivascular	Vessel	Ependymoma

Squamous lesions

	SCC	BCC	NPC
Cells	Polygonal, abundant pink cytoplasm	Basaloid, scant cytoplasm	Large, vesicular, syncytial
Clue	Keratin, bridges	Palisading, clefts	Mixed with lymphocytes
Atypia	More pleomorphic	Less pleomorphic	Prominent nucleoli
Stroma	Desmoplasia	Mucinous / fibromyxoid	Lymphoid

Cervix progression

LSIL (CIN 1)	HSIL (CIN 2–3)	SCC
Koilocytosis; low-grade dysplasia	Loss of maturation to upper 2/3 or full thickness; no invasion	Dysplasia + stromal invasion

Gastric adenocarcinoma

	Intestinal type	Diffuse type
Pattern	Cohesive glands	Single cells, small clusters
Cell	Columnar, elongated nucleus	Round, nucleus pushed aside
Mucin	Apical / luminal	Voluminous intracellular
Gross	Mass ± ulcer	Diffuse wall thickening
Gene	Wnt / APC	CDH1

Spindle cell tumors

	Leiomyoma	Schwannoma	Neurofibroma
Border	Circumscribed	Encapsulated	No capsule
Nuclei	Cigar, blunt	Wavy, tapered	Thin, wavy, comma
Cytoplasm	Pink, fibrillary	Pale, indistinct	Scant
Pattern	Short fascicles, right-angle crossings	Longer fascicles; Antoni A / B, Verocay bodies	Loose; plexiform or diffuse
Stroma	Hyalinization in large tumors	Myxoid (B), hyalinized vessels	Shredded carrot, mast cells

Vascular tumors

	Hemangioma	Lymphangioma	Angiosarcoma
Border	Circumscribed	Dilated channels	Infiltrative
Lumen	Blood	Lymph, lymphocytes	Poorly formed
Lining	Single, uniform	Single, bland	Multilayered, atypical

Melanocytic

	Intradermal nevus	Melanoma
Border	Circumscribed	Infiltrative
Maturation	Present	Absent
Cells	Bland, few mitoses	Pleomorphic, big nucleoli, mitoses

Rejection: acute → chronic

Site	Acute	Chronic
Liver duct	Bile duct injury	Ductopenia
Liver vessel	Endotheliitis	Obliterative arteriopathy
Kidney tubule	Tubulitis	Tubular atrophy
Interstitium	Inflammation	Fibrosis
Arteriole	Endotheliitis	Intimal thickening
Capillaries	Capillaritis, glomerulitis	Basement membrane duplication

Pink material

Fibrosis	Fibrinous	Fibrinoid necrosis	Amyloid
Wavy collagen, fibroblasts	Fibrin strands, inflammation	Bright pink in vessel walls, debris	Glassy, acellular, cracks

Immunological disorder

Amyloidoma (amyloidosis) ★ PA0030 (25)

考古: 11 Salivary gland · 09 Lung · 07 Lung or heart

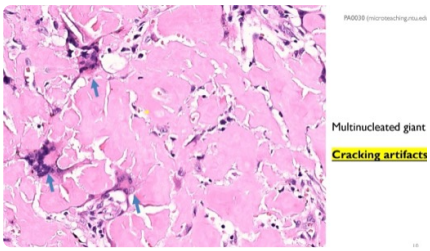
Spot it: Lung replaced by acellular, glassy pink material with cracks; giant cells at the edge.

器官 Lung · **診斷** Amyloidoma (or Amyloidosis)

描述

1. Amorphous, eosinophilic, hyaline extracellular deposits with cracking artifacts
2. Other finding: lymphocytes & multinucleated giant cells (like a foreign-body reaction)

Trap: Not fibrosis (wavy collagen with fibroblasts). Amyloid = Congo red, apple-green birefringence.



Glassy pink amyloid with cracking artifacts; giant cells (arrows)

Chronic (active) rejection PA0284 (69)

考古: not tested in 07–11

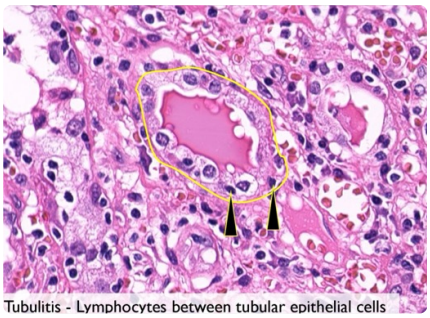
Spot it: Kidney with lymphocytes in tubules, vessels and capillaries, plus atrophy and fibrosis.

器官 Kidney · **診斷** Chronic (active) rejection / chronic active T-cell-mediated rejection

描述

1. Tubulitis, tubular atrophy
2. Interstitial inflammation, interstitial fibrosis
3. Peritubular capillaritis, glomerulitis
4. Endotheliitis, intimal thickening of vessels

Trap: Skipped in class but on the list. Write one acute + one chronic change for each site.



Tubulitis: lymphocytes between tubular epithelial cells (arrowheads)

Acute cellular rejection PA0061 (43); PA0269 (195)

考古: not tested in 07–11

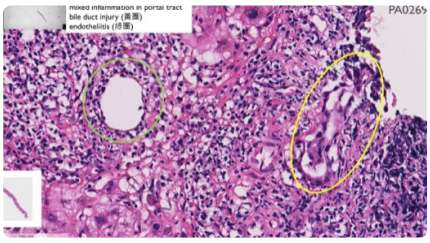
Spot it: Triad in the portal tract: mixed inflammation + bile duct injury + endotheliitis (most specific).

器官 Liver · **診斷** Acute cellular rejection (acute T-cell-mediated rejection)

描述

1. Mixed portal inflammation (lymphocyte predominant, admixed with some neutrophils and eosinophils)
2. Bile duct injury (bile duct epithelium damage and infiltration by inflammatory cells)
3. Subendothelial venous inflammation (endotheliitis) (in severe cases: perivenular parenchymal necrosis)

Trap: Bile duct injury is subtle on PA0061, clearer on PA0269. Scan every portal tract.



PA0269: yellow = bile duct injury; green = endotheliitis; mixed portal inflammation

Allergic rhinitis PA0264 (193)

考古: not tested in 07–11

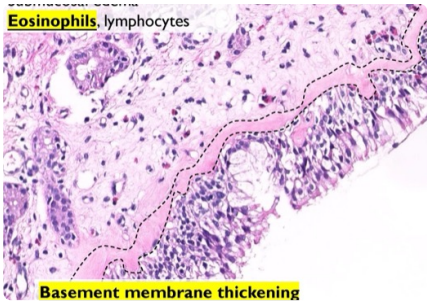
Spot it: Respiratory mucosa with edema and many eosinophils under a thick basement membrane.

器官 Nasal cavity · **診斷** Allergic rhinitis

描述

1. Submucosal eosinophil-rich inflammatory infiltrate
2. Submucosal edema
3. Basement membrane thickening
4. Other finding: squamous metaplasia

Trap: Say eosinophil-rich. Organ = nasal cavity. Squamous metaplasia is an extra finding.



Submucosal edema, eosinophils, thickened basement membrane

Disease of infancy and childhood

Nephroblastoma (Wilms tumor) ★ PA0289 (68)

考古: 11 Kidney · 10 Kidney · 07 Kidney

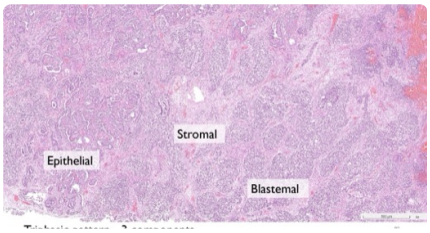
Spot it: Three parts: blue blastema sheets, abortive tubules / glomeruloid structures, spindle stroma.

器官 Kidney · **診斷** Nephroblastoma (or Wilms tumor)

描述

1. Triphasic pattern: (1) blastemal component — sheets of small round blue cells; (2) epithelial component — epithelial cells forming tubular, glomerular or papillary structures
2. Stromal component: spindle cells in a fibromyxoid background (may have heterologous elements — not on this slide)
3. Anaplasia (not on this slide)
4. Necrosis
5. Hemorrhage

Trap: No anaplasia or heterologous elements on this slide — don't write them. Year 10: graded per component.



Triphasic pattern: stromal, epithelial, blastemal

Neuroblastoma PA0305 (107)

考古: 09 Adrenal gland

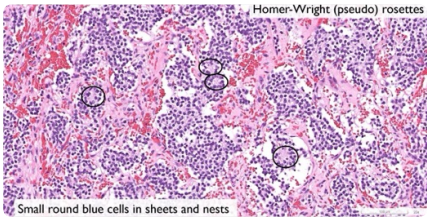
Spot it: Very hemorrhagic adrenal tumor: small round blue cells around pink fibrillary neuropil.

器官 Adrenal gland · **診斷** Neuroblastoma ("poorly differentiated" optional)

描述

1. Circumscribed tumor at the adrenal medulla (border & location)
2. Nests and sheets of neuroblasts: small, round to oval cells with salt-and-pepper chromatin, scant / indiscernible cytoplasm (high N:C ratio) and vague cytoplasmic borders (arrangement & cells)
3. Eosinophilic fibrillary material (neuropil) in the background (thin septa of S100+ schwannian stroma are hard to see — optional)
4. Homer-Wright (pseudo) rosettes (arrangement)
5. Extensive hemorrhage (background)
6. Focal calcification (background)

Trap: Homer-Wright rosette has neuropil in the centre, no lumen. Compressed cortex at the edge = adrenal.



Small round blue cells, Homer-Wright rosettes (circles), pink neuropil

Retinoblastoma PA0247 (142)

考古: 10 Eye 補考

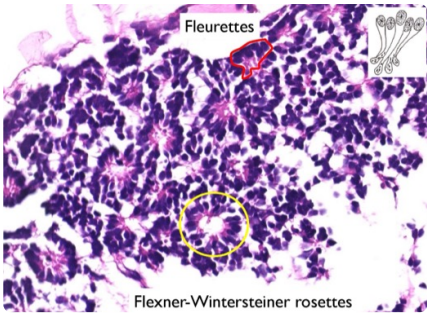
Spot it: Eye: blue tumor from the retina filling the vitreous, rosettes around a true lumen.

器官 Eye · **診斷** Retinoblastoma

描述

1. Small round blue cells arising from the retina (growing into the vitreous cavity without choroidal invasion — for staging)
2. Flexner-Wintersteiner rosettes (rosettes of tumor cells surrounding a central lumen)
3. Fleurettes (photoreceptor differentiation)
4. Necrosis

Trap: Flexner-Wintersteiner = lumen, not neuropil. Choroidal invasion is absent here (staging).



Flexner-Wintersteiner rosettes (circle) and fleurettes

Hyaline membrane disease PA0200 (146)

考古: 08 Lung

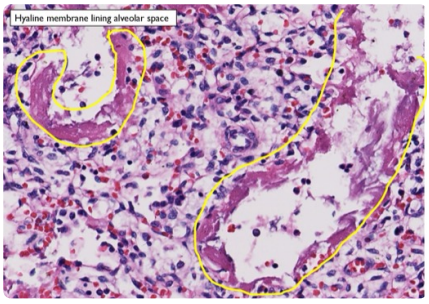
Spot it: Immature lung (cellular cartilage, collapsed alveoli) with pink membranes lining air spaces.

器官 Lung (neonatal) · **診斷** Hyaline membrane disease

描述

1. Variable atelectasis and dilation of alveoli with edematous, thickened alveolar septa
2. Hyaline membrane lining alveolar space: homogeneous eosinophilic linear material closely adherent to the alveolar surface
3. May hemorrhage

Trap: Slide number changed to PA0200. Neonate = hyaline membrane disease; adult = DAD.



Hyaline membranes (outlined) lining air spaces

Epithelial tumor

Squamous cell carcinoma ★ PA0047 (32)

考古: 11 Skin · 10 Uterine cervix · 08 Esophagus · 07 Esophagus

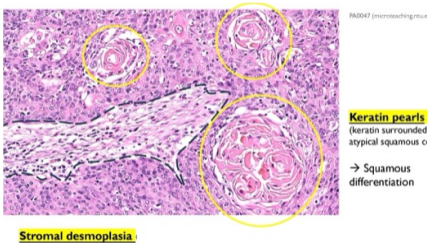
Spot it: Solid nests of polygonal pink cells with keratin pearls and intercellular bridges.

器官 Esophagus · **診斷** Squamous cell carcinoma

描述

1. The tumor cells have pleomorphic hyperchromatic nuclei and prominent nucleoli, arranged in solid sheets & nests (cells & arrangement)
2. The tumor cells are polygonal, with abundant eosinophilic cytoplasm, keratinization and intercellular bridges (differentiation)
3. Invasion to the lamina propria, muscularis mucosae & submucosa (extent of invasion)
4. Desmoplastic stromal reaction (affected background)
5. Squamous dysplasia also present (precursor in the background)

Trap: State how deep it invades, and mention the adjacent squamous dysplasia.



Keratin pearls (circles) and stromal desmoplasia

High-grade squamous intraepithelial lesion (HSIL) PA0337 (CIN 2, online)

· PA0338 (96) (CIN 3)

考古: not tested in 07–11

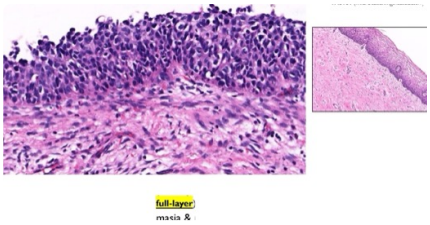
Spot it: Atypical immature squamous cells up to the upper 2/3 or full thickness; basement membrane intact.

器官 Uterine cervix · **診斷** High-grade squamous intraepithelial lesion (HSIL)

描述

1. Atypical squamous cells with nuclear enlargement, hyperchromasia, coarse chromatin, increased N/C ratio, pleomorphism and increased mitosis, expanding to the upper 2/3 of the epithelium / the full thickness of the epithelium
2. Endocervical glandular involvement
3. No stromal invasion

Trap: Round, smooth endocervical gland involvement is NOT invasion. Must write “no stromal invasion”.



Teaching slide PA0101: loss of maturation, atypical cells reaching the upper epithelium

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Squamous cell carcinoma ★ PA0102 (83)

考古: 11 Skin · 10 Uterine cervix · 08 Esophagus · 07 Esophagus

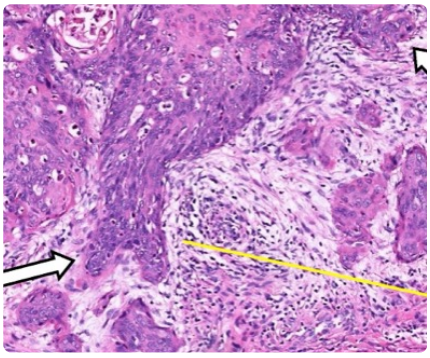
Spot it: Angulated, irregular nests invading cervical stroma with desmoplasia.

器官 Uterine cervix · **診斷** Squamous cell carcinoma

描述

1. Infiltrative, angulated, irregular solid nests of tumor cells with stromal desmoplasia (border, arrangement, background)
2. Tumor cells show pleomorphic hyperchromatic nuclei, increased mitosis, and squamous differentiation: polygonal shape, abundant eosinophilic cytoplasm, keratinization and intercellular bridges (cells, differentiation)
3. Other finding: acute and chronic inflammation with multinucleated giant cells (bystander, not always present)

Trap: HSIL alone scored only 3/5 in year 10. Check for stromal invasion first.



Irregular tumor nests, tongue-like protrusions, stromal desmoplasia

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Basal cell carcinoma PA0295 (118)

考古: not tested in 07–11

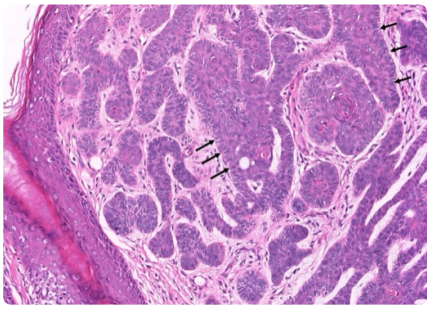
Spot it: Blue basaloid nests from the epidermis with peripheral palisading and retraction clefts.

器官 Skin · **診斷** Basal cell carcinoma (nodular and superficial features)

描述

1. Epidermal tumor with dermal invasion
2. Atypical basaloid tumor cells with elongated hyperchromatic nuclei, arranged in large nests / rounded small nests / irregular jagged nests and cords with peripheral palisading and clefting
3. Increased mitosis and apoptosis
4. Mucinous / fibromyxoid stroma
5. Solar elastosis (background)
6. Perineural invasion (not on this slide)

Trap: Clefts are artifact. Solar elastosis is background. No perineural invasion on this slide.



Basaloid nests with peripheral palisading

Nasopharyngeal carcinoma PA0260 (190)

考古: 11 (M3) Nasopharynx · 10 Nasopharynx 補考

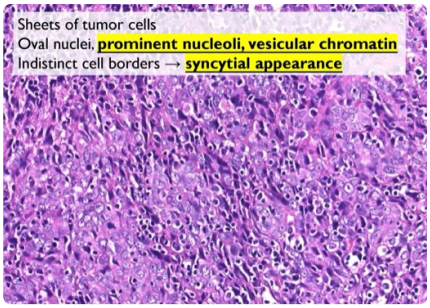
Spot it: Respiratory epithelium + lymphoid stroma with big vesicular nuclei in a syncytium among lymphocytes.

器官 Nasopharynx · **診斷** Nasopharyngeal carcinoma (NPC)

描述

1. Infiltrative growth
2. Sheets of tumor cells mixed with abundant lymphocytes
3. Tumor cells: (1) oval nuclei, prominent nucleoli, vesicular chromatin; (2) indistinct cell borders imparting a syncytial appearance

Trap: No keratinization. Easy to overlook as just lymphoid tissue — look for the large nuclei.



Sheets of vesicular nuclei, prominent nucleoli, syncytial look, lymphocytes

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Familial adenomatous polyposis (multiple tubular adenomas) PA0207 (151)

考古: not tested in 07–11

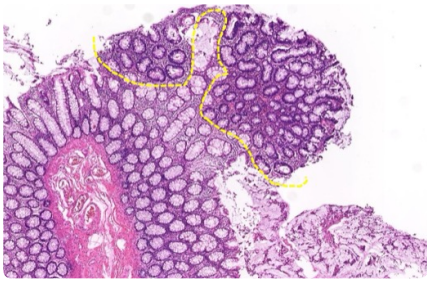
Spot it: Multiple polyps; dark crypts with elongated stratified nuclei next to normal mucosa.

器官 Large intestine OR colon · **診斷** Multiple adenomas, compatible with familial adenomatous polyposis / Tubular adenoma, multiple

描述

1. Multiple polypoid lesions
2. Atypical tall columnar cells with nuclear hyperchromasia, elongation, and stratification, arranged in tubules
3. No stromal invasion

Trap: Diagnosis wording: multiple adenomas, compatible with FAP. Must write “no stromal invasion”.


Adenoma (outlined) with abrupt transition from normal crypts

Adenocarcinoma ★ PA0249 (38)

考古: 11 Large intestine · 10 Large intestine / colon · 09 Colon

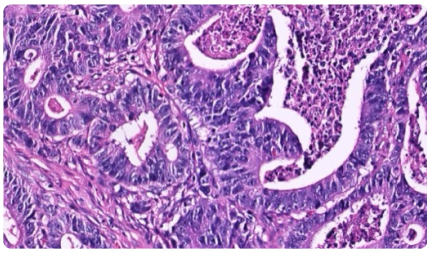
Spot it: Complex or cribriform glands of tall columnar cells with dirty necrosis, invading the wall.

器官 Colon OR large intestine · **診斷** Adenocarcinoma

描述

1. Atypical tall columnar cells arranged in a complex glandular pattern
2. Nuclear hyperchromasia, elongation, and stratification
3. Prominent nucleoli
4. Stromal desmoplasia
5. Dirty necrosis
6. Stromal invasion (to the lamina propria, muscularis mucosae, submucosa and muscularis propria)

Trap: Give the depth of invasion. Year 10: “stomach” as organ scored 0.


Complex glands with dirty necrosis in the lumina

Adenocarcinoma, signet-ring cell type (signet-ring cell carcinoma)

★ PA0279 (33)

考古: 11 Stomach · 10 Stomach · 09 (M3) Stomach · 08 (M3) Stomach

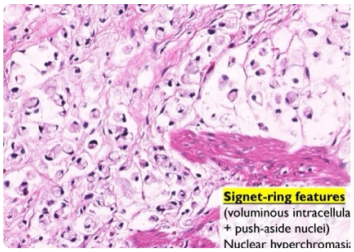
Spot it: Discohesive single cells with a clear mucin globule pushing the nucleus aside.

器官 Stomach (with duodenal involvement) · **診斷** Adenocarcinoma, diffuse/signet-ring cell type (signet-ring cell carcinoma)

描述

1. Infiltrative growth, stromal invasion (deep into the muscularis propria)
2. Small clusters or single discohesive cells with signet-ring features (voluminous intracellular mucin + pushed-aside nuclei)
3. Mild cytologic atypia: nuclear hyperchromasia
4. Stromal desmoplasia

Trap: “Adenocarcinoma” alone = 2/5. Brunner glands mean duodenum, not intestinal metaplasia.



Signet-ring cells: intracellular mucin, nuclei pushed aside

Adenocarcinoma, intestinal type PA0049 (34)

考古: not tested in 07–11

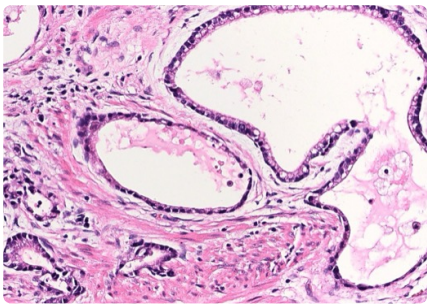
Spot it: Cohesive irregular glands invading the wall, on a background of intestinal metaplasia and dysplasia.

器官 Stomach · **診斷** Adenocarcinoma, intestinal type

描述

1. Infiltrative growth, stromal invasion (penetrates into the subserosal tissue)
2. Irregular glands with focal compact arrangement
3. Flattened, cuboidal to columnar tumor cells with eosinophilic cytoplasm and intracellular mucin
4. Cytologic atypia: nuclear hyperchromasia, occasional prominent nucleoli
5. Stromal desmoplasia
6. Intestinal metaplasia and dysplasia
7. Lymphovascular invasion, perineural invasion (may not be found)

Trap: Labelled PA0049 (34) in the lecture. Always state the type. Atypia may be mild.



Irregular glands; flattened to columnar cells; intracellular mucin

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Neuroendocrine tumor (carcinoid tumor) ★ PA0055 (36)

考古: 11 (M3) Small intestine (duodenum) / pancreas & duodenum · 10 (M3) Pancreas · 09 (M3) Small intestine · 07 (M3) Small intestine

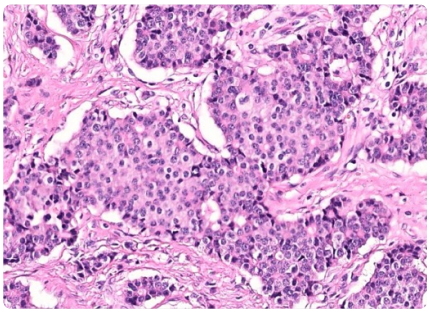
Spot it: Nests, islands and trabeculae of very uniform cells, pink cytoplasm, salt-and-pepper chromatin.

器官 Small intestine · **診斷** Neuroendocrine tumor

描述

1. Infiltrative growth, stromal invasion (deep into the subserosa)
2. Uniform cells with pink cytoplasm, round to oval nuclei, "salt and pepper" chromatin
3. Islands, nests, trabeculae, glands

Trap: Looks bland but infiltrates to the subserosa. Organ was changed in exams (pancreas, duodenum).



Uniform cells with salt-and-pepper chromatin

Non-epithelial tumor

Leiomyoma PA0104 (85)

考古: not tested in 07–11

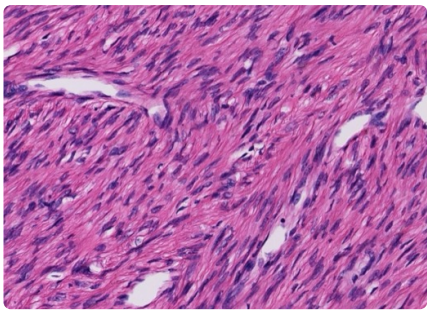
Spot it: Well-circumscribed myometrial tumor of intersecting fascicles with cigar-shaped nuclei.

器官 Uterine corpus · **診斷** Leiomyoma

描述

1. A well-circumscribed tumor composed of uniform spindle cells arranged in intersecting fascicles
2. Tumor cells with cigar-shaped nuclei and eosinophilic cytoplasm
3. No cytologic atypia

Trap: Nuclei are blunt-ended (schwannoma: wavy, tapered). Write “no cytologic atypia”.



Uniform spindle cells, cigar-shaped nuclei, eosinophilic cytoplasm

Schwannoma PA0158 (135)

考古: 11 Soft tissue · 09 Soft tissue

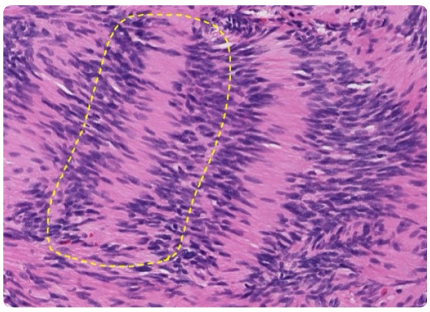
Spot it: Encapsulated tumor alternating cellular Antoni A (Verocay bodies) and myxoid Antoni B.

器官 Soft tissue · **診斷** Schwannoma

描述

1. Well-circumscribed / encapsulated tumor
2. Uniform spindle cells with wavy nuclei
3. Hypercellular (compact) areas with intersecting fascicles and whorling of spindle cells and nuclear palisading, forming Verocay bodies (Antoni A)
4. Hypocellular areas with myxoid stroma (Antoni B)
5. May have hyalinized vessels
6. May have lymphoid aggregates at the periphery
7. May have degenerative changes: scattered bizarre nuclei, cystic change, xanthomatous change

Trap: Bizarre nuclei are degenerative, not malignant.



Antoni A: nuclear palisading forming Verocay bodies

Neurofibroma, plexiform and diffuse type PA0195 (134)

考古: 10 Skin

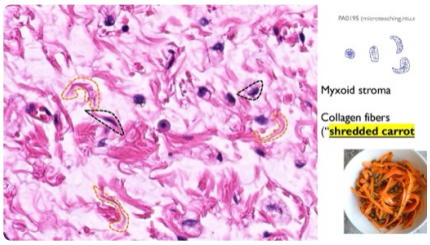
Spot it: Unencapsulated, loose bland spindle cells, shredded-carrot collagen, mast cells.

器官 Soft tissue · **診斷** Neurofibroma, plexiform and diffuse type (the type — localized / plexiform / diffuse — must be written in the diagnosis or description)

描述

1. Infiltrative (diffuse type) or multinodular growth / bag-of-worms appearance (plexiform type)
2. Bland spindle cells with wavy, tapering nuclei
3. Myxoid to collagenous stroma
4. Shredded-carrot-like collagen fibers
5. No cytologic atypia
6. May have entrapped nerves
7. May have mast cell infiltration

Trap: The lecture says the type (localized / plexiform / diffuse) must be written.



Myxoid stroma with shredded-carrot collagen

Hemangioma PA0314 (197)

考古: not tested in 07–11

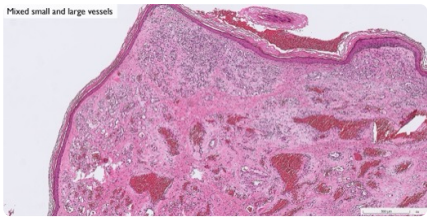
Spot it: Polypoid, circumscribed, vaguely lobular collection of capillaries and dilated blood-filled spaces.

器官 Skin · **診斷** Hemangioma

描述

1. Polypoid lesion composed of variably sized vessels arranged in a vaguely lobular pattern with a circumscribed border
2. Closely packed thin-walled capillaries and dilated irregular vascular spaces
3. Endothelial cells are uniform, flat, single-layered, without cytologic atypia

Trap: Single flat endothelial layer, no atypia (angiosarcoma: multilayered, infiltrative).



Polypoid lesion: mixed small and large vessels

Lymphangioma PA0336 (132)

考古: not tested in 07–11

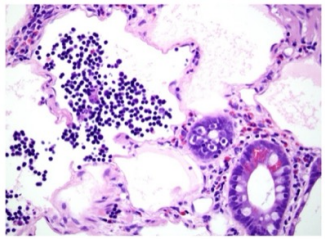
Spot it: Thin-walled dilated channels holding pale lymph and lymphocytes; lymphoid aggregates.

器官 Soft tissue (pericardial) · **診斷** Lymphangioma

描述

1. Aggregation of lymphatic channels with bland-looking endothelial lining
2. Amorphous lymph (proteinaceous fluid) in some spaces, containing lymphocytes and sometimes erythrocytes
3. Lymphocytic aggregates

Trap: The lecture has no photo of PA0336 itself. Check the glass slide.


Similar case (mediastinum), not the exam slide: lymphatic spaces with lymphocytes

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Embryonal rhabdomyosarcoma PA0226 (167)

考古: not tested in 07–11

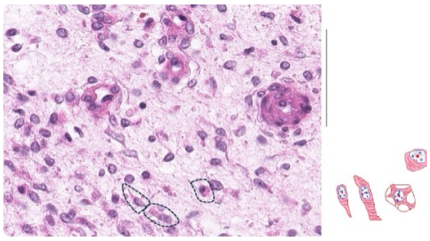
Spot it: Alternating cellular and myxoid areas, cells crowd around vessels; pink rhabdomyoblasts.

器官 Soft tissue · **診斷** Embryonal rhabdomyosarcoma

描述

1. (Infiltrative tumor border — cannot be assessed on this slide)
2. Hypercellular & hypocellular areas, higher cellularity around blood vessels
3. Primitive spindle and round to stellate cells with eosinophilic cytoplasm, eccentric nuclei, and cross striation
4. Myxoid stroma

Trap: The tumor border can't be assessed on this slide. Look for cross-striations at high power.


Primitive cells; eosinophilic cytoplasm, eccentric nuclei

Malignant melanoma PA0100 (81)

考古: 10 (M3) Lung

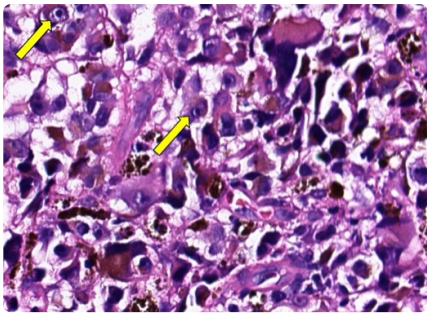
Spot it: Very cellular tumor in epidermis and dermis; big nucleoli, melanin, many mitoses.

器官 Skin · **診斷** Melanoma

描述

1. Infiltrative tumor involving the epidermis and dermis
2. Highly cellular tumor composed of irregular nests or single cells with enlarged pleomorphic nuclei, prominent nucleoli, and melanin pigments
3. High mitotic activity
4. Brisk tumor-infiltrating lymphocytes

Trap: Unlike a nevus: no maturation, marked atypia, mitoses. Metastasis to lung appeared in year 10.



Pleomorphic nuclei, prominent nucleoli, melanin

Mature (cystic) teratoma PA0285 (88)

考古: 10 Ovary 牙, 補考 · 07 Ovary

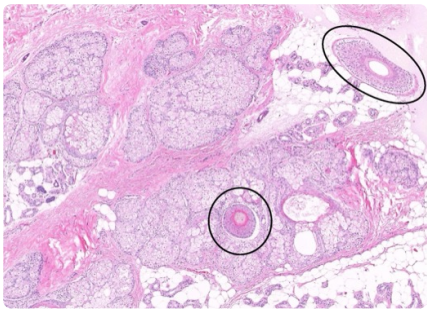
Spot it: Cyst with mature tissues from different germ layers: skin, fat, cartilage, glia, respiratory epithelium.

器官 Ovary · **診斷** Mature teratoma

描述

1. Cystic tumor composed of mature tissue from different germ layers, including squamous epithelium, skin adnexa, glial tissue, nerves, adipose tissue, bone, cartilage, respiratory epithelium, and choroid plexus (in the exam, list the tissues you actually see on the slide)

Trap: List only what you see on your slide. “Teratoma” alone = 3/5; each tissue 2–3 points (year 10).



Skin, adnexa, hair shafts and fat in the cyst wall

Cardiovascular diseases

Cardiac myxoma PA0002 (5)

考古: 08 Heart (left atrium), 68F

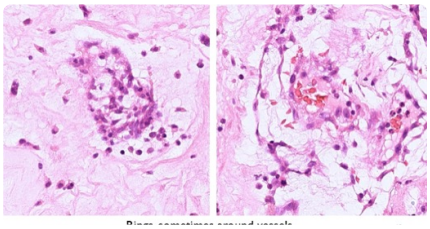
Spot it: Pale myxoid tumor with single cells, cords and rings; hemorrhage and hemosiderin.

器官 Heart (no need to name the chamber) · **診斷** Cardiac myxoma or Myxoma

描述

1. Well-circumscribed tumor
2. Bland spindle or stellate cells with oval nuclei arranged singly or in cords and rings
3. Myxoid stroma
4. Abundant vessels
5. Hemorrhage and hemosiderin

Trap: “Heart” is enough; no need to name the chamber.



Myxoma cells forming rings, sometimes around vessels

Atherosclerosis ★ PA0170 (2)

考古: 10 Artery · 09 Artery · 07 Artery

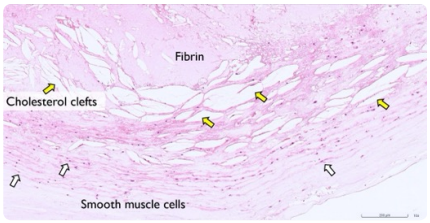
Spot it: Thick intima: fibrous cap over a necrotic core with needle-shaped clefts and foam cells.

器官 Artery (no need to name which artery) · **診斷** Atherosclerosis

描述

1. Fibrous cap with smooth muscle cells and collagen
2. Necrotic core with cholesterol clefts and foam cells
3. Macrophages and lymphocytes

Trap: “Artery” = 2; “blood vessel” or “aorta” = 1 (year 10). Lecture photos are labelled PA0170.



Cholesterol clefts, fibrin, smooth muscle cells

Acute rheumatic pancarditis PA0197 (143)

考古: not tested in 07–11

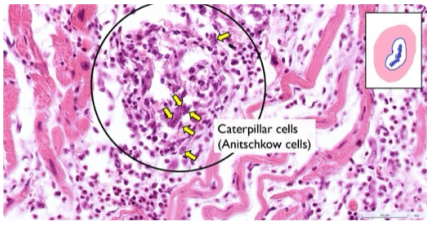
Spot it: All three heart layers inflamed; Aschoff nodules with wavy-chromatin caterpillar cells.

器官 Heart · **診斷** Acute rheumatic pancarditis

描述

1. Acute inflammation involving the pericardium, myocardium and endocardium
2. Aschoff nodules composed of numerous lymphocytes and histiocytes, some forming multinucleated giant cells (Aschoff giant cells), admixed with Anitschkow cells

Trap: Name all three layers: pericarditis (fibrinous), myocarditis, endocarditis.



Aschoff body with caterpillar (Anitschkow) cells

Amyloidosis ★ PA0171 (3)

考古: 11 Salivary gland · 09 Lung · 07 Lung or heart

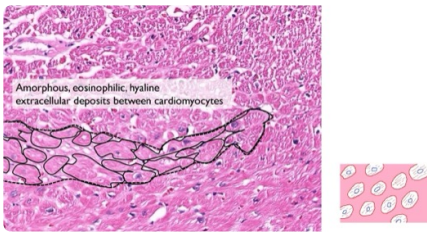
Spot it: Pale, glassy material around myocytes and in vessel walls, separating the fibres.

器官 Heart · **診斷** Amyloidosis

描述

1. Amorphous, eosinophilic, hyaline extracellular deposits
2. Deposition patterns: myocardial interstitial (pericellular / nodular); vascular / microvascular; also deposits in the endocardium and epicardium

Trap: Describe the pattern: pericellular / nodular, vascular. Same material as the lung amyloidoma.



Glassy deposits between cardiomyocytes (outlined)
